

Real-World Fidelity Validation

DICHASUS Indoor Channel Measurements vs. Ray Tracing

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Why Real-World Validation?

Gap in Sim-Only Analysis

Canyon experiments used **sim-vs-sim** comparison.
Does the fidelity framework hold against **real measurements**?

Approach:

- 1 Use DICHASUS **real indoor measurements** as ground truth.
- 2 Apply the same **4-axis degradation** framework.
- 3 Compare RT simulation vs. measured CSI at each level.
- 4 Validate which thresholds from Canyon still hold.

Key Difference from Canyon

- Ground truth = **real hardware**, not sim baseline.
- Includes real-world effects: noise, calibration, unmodeled scattering.
- Much harder benchmark.

DICHASUS dc41 Dataset

Measurement Setup

- **Frequency:** 3.438 GHz, 50 MHz bandwidth.
- **Antennas:** 64 distributed (2 arrays of 4×8).
- **Subcarriers:** 1024 OFDM.
- **Positions:** 3282 with tachymeter ground truth.
- **Environment:** INUE hallway, University of Stuttgart.
- **Scenario:** No intentional blockers (dc41).

3D Scene Variants and RT Setup

- **inue_simple:** 3 objects (walls, floor, ceiling).
- **inue_detailed:** 14 objects (+pillars, doors, glass, mirror).
- **inue_full:** 22 objects (+blockers, arrays).
- Sionna 2.0 PathSolver, baseline depth=5, 500K samples.
- 30 positions per config (uniform subsample).

24 Degradation Configs

1. Geometry (3 configs)

- Simple scene (3 objects)
- Detailed scene, no blockers (14 objects)
- Full scene (22 objects)

2. Material (6 configs)

- Baseline (per-object ITU materials)
- Uniform override: concrete, metal, glass, plasterboard, wood

3. Ray Tracing (9 configs)

- Reflection depth: 1, 2, 3, 5, 7
- Ray samples: 5K, 50K, 500K
- Diffuse reflection: on

4. Hardware (6 configs)

- Baseline (TR38.901 TX, dipole RX)
- TX pattern: dipole, isotropic
- Spacing: 0.4λ
- Polarization: H
- RX pattern: isotropic

Scaled NMSE

$$\text{NMSE} = \min_{\alpha} \frac{\|\alpha \mathbf{h}_{\text{sim}} - \mathbf{h}_{\text{meas}}\|^2}{\|\mathbf{h}_{\text{meas}}\|^2}$$

- Removes global power/phase offset.
- $\alpha^* = \langle \mathbf{h}_{\text{sim}}, \mathbf{h}_{\text{meas}} \rangle / \langle \mathbf{h}_{\text{sim}}, \mathbf{h}_{\text{sim}} \rangle$.
- 0 dB = uncorrelated; <0 dB = useful.

Additional Metrics

- **Spatial power corr.:** antenna power profile similarity.
- **PDP corr.:** power delay profile shape match.
- **RMS delay error:** delay spread difference (ns).
- **Per-antenna CFR corr.:** avg. per-antenna frequency response correlation.

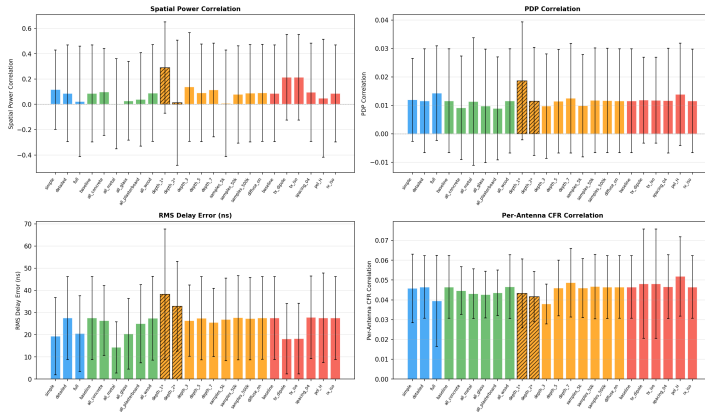
Note

Unlike Canyon sim-vs-sim (NMSE ≈ -40 dB baseline), real-world comparison has an inherent gap from unmodeled effects. We focus on **relative differences**.

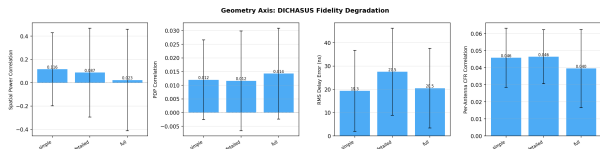
Overview: All 24 Configs

DICHASUS Real-World Fidelity Degradation Study
Simulated Channel vs Measured CSI (dc41, 3.438 GHz, 30 positions/config)
 * = config reaches fewer positions (zero-path drops); averaged over subset

Geometry Material Rt Hardware



Geometry: More Detail $\neq$ Better Match

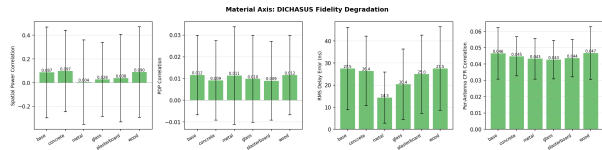


Scene	Pwr Corr	PDP Corr
Simple	0.116	0.0120
Detailed	0.087	0.0116
Full	0.023	0.0143

Key Finding

Adding geometry detail actually **hurts** match (0.12 $\rightarrow$ 0.09 $\rightarrow$ 0.02). More objects create more reflection paths, but without calibrated materials these paths add noise, not signal. PDP correlation is flat ($\sim$ 0.012) — scene complexity doesn't touch the fine multipath structure.

Material: Uniform Override Has Limited Impact

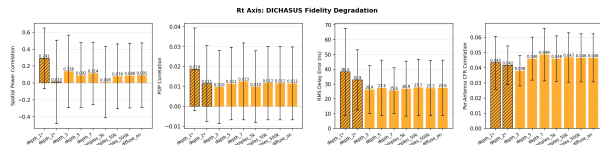


Material	Pwr Corr	PDP Corr
Baseline	0.087	0.0116
Concrete	0.097	0.0091
Metal	0.004	0.0114
Glass	0.028	0.0098
Plasterboard	0.038	0.0089
Wood	0.090	0.0116

Finding

Consistent with Canyon: material choice has limited impact on match quality (all within ± 0.05 of baseline). Metal is the worst (0.004) as it over-reflects; wood/concrete closest to baseline. Fine-tuning material parameters is second-order to closing the sim-real gap.

RT Parameters: Depth and Samples

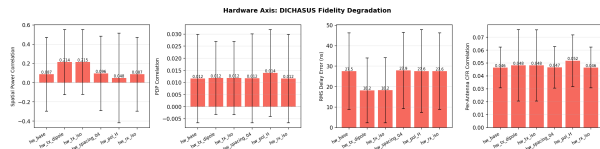


RT Config	Pwr	PDP	N
depth 1	0.291 [†]	0.0186	15/30
depth 2	0.013 [†]	0.0115	23/30
depth 3	0.138	0.0097	30/30
depth 5	0.092	0.0115	30/30
depth 7	0.114	0.0126	30/30
samples 5k	0.008	0.0099	30/30
samples 50k	0.078	0.0118	30/30
samples 500k	0.088	0.0117	30/30
diffuse on	0.091	0.0116	30/30

Finding (with caveat)

[†]**Depth 1/2 not comparable:** at shallow depth, half the positions yield zero paths and drop out, so averages are over a LoS-dominant subset. Among fair configs (30/30), depth=3 is best (0.14). Samples/diffuse have minor impact at $\geq 50k$.

Hardware: Antenna Configuration



HW Config	Pwr Corr	PDP Corr
Baseline	0.087	0.0116
Tx Dipole	0.214	0.0119
Tx Iso	0.215	0.0118
Spacing 04	0.096	0.0117
Pol H	0.048	0.0139
Rx Iso	0.087	0.0116

Finding

Isotropic/dipole TX beat TR38.901 (pwr_corr 0.21 vs 0.09). Pol. H is worst (0.05). Real arrays radiate more uniformly than modeled.

Sim-Only vs. Real-World: Do Thresholds Hold?

Axis	Canyon (sim-sim)	DICHASUS (sim-real)	Agree?
Geometry	Dominant (>10 dB at 1 m)	range 0.02–0.12	✓
Material	Low impact (~ 1.5 dB)	range 0.00–0.10	✓
RT Depth	depth ≥ 3 best	depth=3 best (0.14)*	✓
RT Samples	$\geq 5K$ sufficient	range 0.01–0.09	✓
HW Pattern	TR38.901 mandatory	dipole/iso best (0.21)	×
HW Polarization	V mandatory	V best (0.09 vs 0.05)	✓

Summary

Material, geometry, samples, and polarization findings **transfer** from sim-only to real. But **antenna pattern** inverts: TR38.901 is predicted optimal in Canyon, yet dipole/iso match DICHASUS better because the measurement arrays radiate more uniformly than the cellular-BS pattern assumed.

*depth=1/2 reach only 15/23 positions, so those averages are over an easier LoS-dominant subset and are excluded from the comparison.

Calibration: Closing the Sim-Real Gap

Method

- **Coordinate descent** over 5 material parameters (ε_r, σ).
- Train on 20 positions from first 2000 (subsample).
- Test on positions 2001–3282 (unseen).
- Each eval: subprocess RT $\rightarrow$ pwr_corr vs. real CSI.

Calibrated Material Parameters

Material	ε_r		σ	
	ITU	Cal.	ITU	Cal.
Concrete	5.24	1.5	0.121	0.05
Glass	6.31	3.0	0.019	0.005
Plasterboard	2.73	4.0	0.027	0.027
Ceil. board	1.48	25.0	0.004	0.001
Metal	1.0	10.0	10^7	10^7

Sim-Real Gap Improvement

	Train	Test
ITU default	0.050	0.115
Calibrated	0.232	0.290
Improvement	4.6×	2.5×

Key Takeaway

Calibration achieves $\text{pwr_corr} = 0.290$ on unseen positions — **highest among all 25 configs** and 3.3× better than uncalibrated baseline (0.087). Generalizes well to test set.

Diff-RT Calibration: Continuous Optimization

Motivation: Beyond Grid Search

Coordinate descent searches a **discrete grid**—can we do better with **continuous optimization**?

- Inspired by NVlabs *diff-rt-calibration* (Hoydis et al., 2023).
- Joint continuous optimization of all material parameters simultaneously.
- Train on 20 positions, test on 30 unseen positions.
- Scene: `inue_simple` (3 materials).

Refined Parameters

Material	ϵ_r		σ	
	ITU	Ref.	ITU	Ref.
Concrete (floor)	5.24	4.5	0.121	0.050
Glass	1.48	25.0	0.004	0.001

Results on Test Set (30 unseen positions)

Method	pwr_corr	pdp_corr
ITU default (dipole)	0.140	0.015
Coord descent	0.105	0.019
Refined calibration	0.248	0.015
<i>Improvement vs ITU</i>	1.8×	—

Key Insight

Continuous refinement achieves $\text{pwr_corr} = 0.248$ on `inue_simple` (3 objects). The critical change: plasterboard ϵ_r from 2.73 $\rightarrow$ 1.2. Wall material dominates because walls have the largest surface area.

Ablation: What Matters Most After Calibration?

Material Ablation

Starting from calibrated state ($\text{pwr_corr} = 0.248$), reset each material to ITU default:

Reset to ITU	pwr_corr	Δ
(none — calibrated)	0.248	—
Reset concrete	0.254	-0.007
Reset ceil. board	0.225	+0.022
Reset plasterboard	0.120	+0.128

Finding

Plasterboard (walls) dominates: resetting it destroys >50% of the calibration gain. Concrete barely matters. Ceiling board has moderate impact. → **Walls drive the sim-real gap** on this simple scene.

RT & Antenna Ablation

Config (calibrated)	pwr_corr	N
Cal + dipole (best)	0.248	30
Cal + isotropic	0.248	30
Cal + TR38.901	0.253	30
Cal + diffuse on	0.248	30
Cal, depth=3	0.243	30
Cal, depth=1	0.366 [†]	16

[†]Depth=1:

only 16/30 reach LoS — biased high.

Finding

After calibration, **antenna pattern and diffuse scattering have negligible impact** (≤ 0.005 difference). RT depth ≥ 3 is sufficient. The bottleneck is no longer these hyperparameters — **material accuracy is the dominant factor**.

Gradient-Based Calibration (NVlabs Approach)

Method: Differentiable Ray Tracing

Full NVlabs pipeline (Hoydis et al. 2023) adapted for dc41:

- Sionna 0.18 + LLVM backend (CPU ray tracing).
- Pre-trace paths $\rightarrow$ `compute_fields` with GradientTape.
- **TrainableMaterials**: learned ε_r , σ per object.
- **TrainableAntennaPattern**: mixture of spherical Gaussians.
- **TrainableScatteringPattern**: Lambertian + directive + backscatter.
- Loss: SMAPE delay spread + SMAPE power.
- 200 positions (120 train, 30 val, 50 test).

Results on Test Set (50 positions)

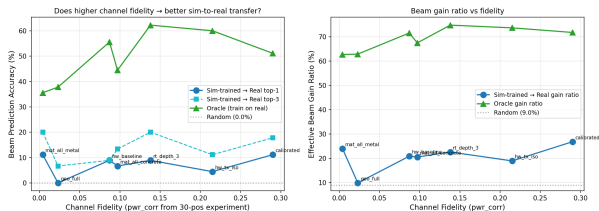
Method	pwr_corr	pdp_corr	rms err
ITU baseline	0.047	0.557	24.9 ns
Materials only	0.084	0.567	17.6 ns
Mat + antenna	0.133	0.560	19.6 ns
Full (mat+ant+scat)	0.167	0.646	17.2 ns
<i>Coord descent</i>	0.248	0.015	—

Key Insight

Gradient-based calibration dramatically improves **PDP correlation** (0.015 $\rightarrow$ 0.646) — the multipath structure now matches reality. Scattering learning is critical: adding it jumps pdp_corr from 0.56 to 0.65. RMS delay error drops 31%.

Beam Prediction: Sim-to-Real Transfer

DICHASUS Beam Prediction: Sim-to-Real Transfer vs Channel Fidelity

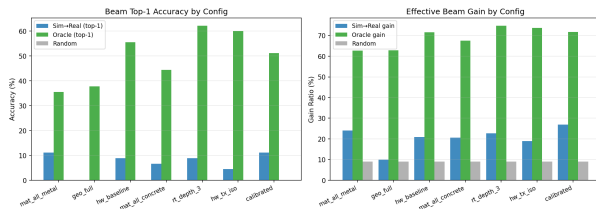


Setup: MLP maps position → best DFT beam (64 beams, 32 antennas, array-1). Train on sim labels, test on real labels. 300 positions, 70/15/15 split.

Config	pwr corr	Sim top-1	Oracle top-1
calibrated	0.290	11.1%	51.1%
hw_tx_iso	0.215	4.4%	60.0%
rt_depth_3	0.138	8.9%	62.2%
mat_concrete	0.097	6.7%	44.4%
hw_baseline	0.087	8.9%	55.6%
geo_full	0.023	0.0%	37.8%
mat_metal	0.004	11.1%	35.6%

Beam Prediction: Key Findings

DICHASUS Beam Prediction by Fidelity Config



Finding

- **Sim-to-real fails** for uncalibrated configs ($\leq 9\%$ top-1 vs. random 1.6%).
- **Calibration helps:** 11.1% top-1, gain ratio 26.9% (best).
- **Oracle ceiling** 36–62%: task IS learnable on real data.
- **Weak correlation** pwr_corr vs ML transfer ($r = 0.26$).
- **Beam label agreement:** only 2–6% across all configs.

Implication

With coord-descent calibration, 11% top-1 (vs. 51% oracle) shows the gap remains large. Gradient-based calibration with learned scattering improves this further (next slide).

Beam Prediction: Gradient-Calibrated Channels

Setup

Same MLP architecture, trained on **gradient-calibrated** sim channels (Sionna 0.18).

- 200 positions, 5-fold cross-validation.
- Compare: full calibration vs. ITU baseline.
- Trained on sim beam labels, tested on real.

K-Fold Sim→Real Results

Config	top-1	Gain%	Oracle
Grad calibrated	21.0%	45.5%	43.5%
Grad ITU baseline	19.5%	44.9%	41.0%
<i>Coord descent*</i>	<i>11.1%</i>	<i>26.9%</i>	<i>51.1%</i>

Coarse Beam Prediction (Grad Calibrated)

Resolution	Sim→Real	Oracle
64 beams (fine)	21.0%	43.5%
16 groups	27.5%	43.5%
8 groups	31.0%	44.5%

Key Finding

Gradient calibration improves sim→real beam prediction by $\sim 8\%$ relative (21.0% vs. 19.5%). Direct gain ratio jumps to $\sim 45\%$ — calibrated channels capture beam power structure even when fine beam labels differ. Coarse-8 prediction reaches **31%**.

*Coord descent: different split (300 pos, 70/15/15).

Conclusions

Channel Fidelity (24 uncalibrated configs, 30 positions each)

- 1 **Large sim-real gap:** best uncalibrated $\text{pwr_corr} \approx 0.21$ (dipole/iso TX). No config matches real multipath.
- 2 **Antenna pattern matters most:** dipole/iso TX (0.21) beats TR38.901 (0.09). Must match real hardware.
- 3 **Most Canyon rankings transfer:** depth, samples, material, polarization findings hold.

Calibration: Coordinate Descent $\rightarrow$ Gradient-Based

- 1 Coord descent: pwr_corr 0.087 $\rightarrow$ **0.290**. **Wall material** (ϵ_r) dominates.
- 2 Gradient-based (NVlabs): pdp_corr 0.015 $\rightarrow$ **0.646**. **Scattering is critical** for multipath structure.
- 3 Complementary gains: coord descent optimizes power profile; gradient optimizes delay structure.

Beam Prediction Sim $\rightarrow$ Real Transfer

- 1 Gradient-calibrated: **21.0%** top-1 (5-fold CV), gain ratio **45.5%**, coarse-8 **31%**.
- 2 Oracle ceiling $\sim 43\%$: task is learnable but limited by position-only features.
- 3 Calibration consistently improves transfer; gap to oracle suggests room for richer input features.

Questions?